A Reversible Generative Network for Hyperspectral Unmixing With Spectral Variability

Yuyou Gao, Bin Pan, *Member, IEEE*, Xia Xu, Xinyu Song, and Zhenwei Shi, *Senior Member, IEEE*

***Abstract*— Spectral variability is one of the challenges for hyperspectral unmixing. Recently, deep generative models are developed to describe the spectral variability, which have attracted increasing attention. However, generative unmixing methods may suffer the problems of mode collapse and image blur, which tend to generate uncontrollable endmember distribution. To address this issue, in this article, we propose a reversible generative network (Rev-Net) for hyperspectral imagery unmixing, which targets at the spectral variability challenge. Our motivation is that if the endmember distribution can be described by an explicit mathematical expression and the expression is reversible, then the generation process will be more stable. To achieve this purpose, Rev-Net mainly includes two contributions: a flow-based endmember learning module and a theoretical proof for the reversibility of the endmember generation process. In the endmember learning module, we develop a new flow-based structure with a series of reversible transformation, so as to obtain an explicit mathematical expression for the endmember distribution. Moreover, to guarantee the existence of the explicit expression, we have theoretically proven the reversibility of the endmember learning module. Through the flow-based endmember learning module and the correspond theoretical analysis, the proposed Rev-Net can make the endmember generation process more stable and thus avoiding the problems of mode collapse and image blur. In addition, we also construct an abundance guidance module to further assist in the generation process of endmember by image reconstruction. Experimental results on real hyperspectral datasets and synthetic datasets indicate that Rev-Net has certain competitiveness. The codes are available at https://github.com/Lab-PANbin/Rev-Net.**

***Index Terms*— Generative model, hyperspectral unmixing, reversible neural networks, spectral variability.**

I. Introduction

HYPERSPECTRAL remote sensing shows strong ability in detecting ground object information, and the common applications include target detection and hyperspectral image classification. However, due to the high complexity of the natural surface and the limitation of technology, hyperspectral data are composed of mixed pixels and may not be utilized directly. For the better application of hyperspectral remote sensing data, we usually use unmixing technology, which aims at decomposing the mixed pixels into the spectral information of the pure pixels (endmembers) and their proportion (abundance) [1], [2].

Hyperspectral unmixing has two main tasks, which are endmember extraction and abundance estimation, respectively. The unmixing models can be roughly divided into two kinds, namely linear mixing model (LMM) [3], [4] and nonlinear mixing model [5]. Traditional unmixing methods can be divided into geometrical methods [6], [7], [8], statistical based approaches [9], [10], sparse regression methods [11], [12], and nonnegative matrix factorization (NMF) [2], [13]. Deep learning has shown great potential in various fields, so hyperspectral unmixing based on deep learning has also been developed rapidly in recent years [14], [15], [16]. Autoencoder [1], [17] is one of the most commonly used forms. Researchers have made many improvements to end-to-end autoencoder unmixing network, such as either multitask [18] and multistream [19], [20] unmixing through multiple autoencoders or improving loss function [21], [22].

However, the method mentioned above may not handle the spectral variability of real remote sensing scenarios, which limits the effectiveness of unmixing algorithms. Spectral variability is an important factor that researchers need to consider in hyperspectral unmixing tasks, which means that the spectral characteristics of pure endmembers will change in the observed hyperspectral image [23]. There are many reasons for this transformation, which can be classified as external and internal factors. The external reasons are the change in lighting or the difference in atmospheric conditions. The internal reasons are the intrinsic characteristics of pure materials. To address this problem, researchers have proposed several theories to simulate spectral variability [24], [25], [26], [27].

At present, the popular methods to consider spectral variability can be divided in two classes. On the one hand, researchers have proposed some ideas increasing robustness and thus avoiding the influence of spectral variability on the results of unmixing. On the other hand, researchers have changed the assumptions of the model and defined some functions to make the endmembers used in the reconstruction

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of hyperspectral images have some variability. Spectral library methods involve multiple endmember spectral mixture analysis [28] and sparse spectral unmixing [29]. The unmixing methods using Bayes-based can capture endmember variability through Bayesian posterior distribution, such as beta compositional model [30] and Gaussian mixture models [31]. Some parameter EM methods [32], [33], [34] consider using a function to represent the variability of endmembers as well as obtain the abundance and parameters used in the function by solving the optimization problem. Other EM methods that do not consider models, such as augmented linear mixing model [35] and RUSAL [36], increase robustness and indirectly reduce spectral variability by regularization. However, the methods taking into account spectral variability may not take advantage of deep learning in mining information and fitting unknown models about endmember changes. Therefore, with the development of generative models in the field of deep learning, researchers have proposed the idea which applies generative models to the extraction of endmembers with spectral variability.

In recent years, generative models have been increasingly employed for conducting the task of hyperspectral unmixing considering spectral variability. Different from those traditional unmixing methods considering spectral variability, researchers have proposed various approaches to autonomously learn the distribution of endmembers. Those methods take advantage of deep learning methods to address spectral variability. Specifically, some deep generative neural networks have been utilized to model the distribution of endmembers. Current generation models [37], [38], [39], [40], [41] such as variational autoencoders (VAE) and generative adversarial networks (GAN) have been utilized and demonstrated excellent performance. The characteristics of the generative model enable us to describe the variability of endmember. We treat the endmember at each pixel as a sample from the distribution. This distribution is regarded as endmember which has a definite but unknown variability. This idea avoids making assumptions about the form of spectral variation or artificially adding restrictions.

However, existing generative models for unmixing may suffer the problems of mode collapse and image blurring [42], which indicates an unstable training process and limiting the ability of algorithms in unmixing task. Specifically, the algorithm using VAE cannot directly optimize the objective function, and usually approximates the objective function by making the variational distribution more similar to the prior distribution. The prediction performance of VAE may be biased due to the existence of KL divergence, which results in inaccurate endmember generation. The algorithm with GAN needs to reach Nash equilibrium in the training process. Otherwise, the problem of mode collapse will occur, which will affect the acquisition of the endmember distribution. Therefore, we consider to overcome these limitations of these generative unmixing models with a flow-based model [43].

We propose an unmixing model for spectral variability based on reversible neural networks. The whole model is composed of endmember learning module and abundance guidance module. In the endmember learning module, we can specify the mathematical form of the endmember distribution. In order to learn a proper endmember distribution, we introduce a reversible neural network, in which the endmember generation process is controllable and stable, and the input information of hyperspectral image is fully utilized. Furthermore, a theoretical analysis is presented to demonstrate the reversibility of the endmember learning module. The main contributions of our work are as follows.

1) We propose a reversible generative network to address the spectral variability problem in hyperspectral unmixing, which aims at providing a more stable endmember generation process.
2) We construct a reversible module for endmember learning, which describes the endmember distribution by explicit mathematical expressions.
3) We have theoretically proven the reversibility of the endmember learning module so as to guarantee the effectiveness of the proposed method.

Section II-A expresses the linear mixed model considering endmember variability. The overall framework of the model is introduced in Section II-B. In Sections II-C and II-D, the endmember learning module and abundance guidance module are introduced, respectively. We provide the proof of reversibility and the total loss function of the model in Sections II-E and II-F. Finally, experimental results on both real datasets and synthetic datasets are shown in Section III, demonstrating the effectiveness of algorithm.

II. Methodology

A. LMM With Spectral Variability

LMM is a widely used mixing model, which is denoted by

$$\mathbf{y}_n = \mathbf{M}_0 \mathbf{a}_n + \mathbf{e}_n \tag{1}$$

for $n = 1, \ldots, N$, where $\mathbf{y}_n \in R^{L\times 1}$ denotes the nth mixed pixel with L spectral bands in hyperspectral image. $\mathbf{M}_0 = [\mathbf{m}_{0,1}, \ldots, \mathbf{m}_{0,P}] \in R^{L\times P}$ is the endmember matrix of the image, which indicates the number of endmembers is P, $\mathbf{e}_n \in R^{L\times 1}$ is the additive noise vector, and $\mathbf{a}_n \in R^{P\times 1}$ is the abundance vector corresponding to the nth pixel. Besides, the abundance vector should satisfy sum-to-one constraint (ASC) and nonnegative constraint (ANC), which is

$$\mathbf{1}_P^T \mathbf{a}_n = 1 \tag{2}$$

$$\mathbf{a}_n \succeq 0 \tag{3}$$

where $\mathbf{a}_n \succeq 0$ denotes each component of $\mathbf{a}_n$ is great than or equal to zero.

LMM assumes that pure endmembers of all pixels are the same. This hypothesis makes the model more concise and the calculation simpler. In many cases, it may affect the accuracy of the endmembers and abundance maps estimated by unmixing algorithm due to the spectral variability of the endmembers.

Due to the influence of light, terrain, atmospheric conditions, and the inherent variability of the material, the spectral characteristics of a certain material will vary. This results in that performance of the unmixing algorithm based on LMM is

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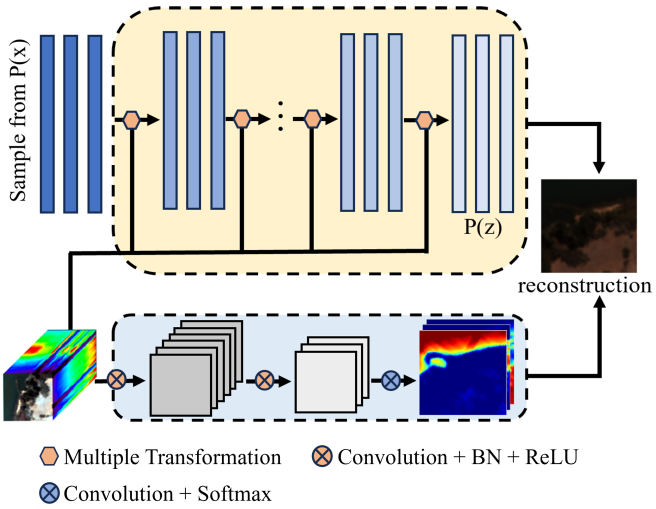


Fig. 1. Overall structure of reversible generative network unmixing model. The yellow part indicates the endmember learning module, and the blue part indicates the abundance guidance module. Multiple transformation includes normalization, permutation operation, and affine coupled transformation.

limited. In this case, using the same endmember matrix M_0 for all pixels in the LMM will produce a large error, resulting in a bias in the estimate of the endmember and the abundance. Due to the significant influence of endmember variability on hyperspectral unmixing, the researchers proposed abundant algorithms that can still be reliable in this case.

Therefore, in the hyperspectral unmixing problem considering spectral variability, we usually believe that there is endmember transformation for each pixel. The mathematical form is expressed as follows:

$$\mathbf{y}_n = \mathbf{M}_n \mathbf{a}_n + \mathbf{e}_n, \text{ subject to } \mathbf{1}_P^T \mathbf{a}_n = 1, \mathbf{a}_n \succeq 0 \qquad (4)$$

for $n = 1, \ldots, N$ where $\mathbf{M}_n \in R^{L \times P}$ denotes the endmember matrix of the nth pixel.

B. Overall Structure of Rev-Net

In this section, we propose the overall structure for reversible generative network unmixing algorithm (illustrated in Fig. 1). The structure of the variable spectral unmixing model based on reversible neural network involves two parts. The first part is the endmember learning module and the second part is the abundance guidance module.

In the endmember learning module, the input vector is obtained by random sampling from the standard normal distribution. The number and dimension of vectors is consistent with the number and dimension of endmembers in the hyperspectral image. The input vector is passed into a reversible neural network, and the estimated endmember vector is outputted after several iterations. Finally, we can obtain an explicit expression of the distribution function and the endmember samples taken during the training process. We ensure that the input vector and the output vector can form a one-to-one mapping relationship, which means that the standard normal distribution and the distribution of endmember with spectral variability can be mutually transformed. In the abundance guidance module, we adopt the convolutional coding structure. Following the encoding structure of the autoencoder unmixing algorithm, we obtain a low-dimensional representations of a hyperspectral image by using multiple convolution layers, which are considered to be the abundance.

The two modules are not completely independent. In the process of endmember generation, hyperspectral images will be added to the training process of the reversible module as known information and guide the generation of endmember matrix. Moreover, for the generated endmember and abundance matrix, the reconstructed image is generated by LMM with spectral variability shown as (4), and the reconstruction loss is constructed with the original hyperspectral image. The training of the endmember guided by abundance is achieved by minimizing the reconstruction loss.

Algorithm 1 Reversible Neural Network for Hyperspectral Unmixing with Spectral Variability

Input hyperspectral image $\mathbf{X}$, sample vector from Gaussian distribution $\mathbf{z}$.
Endmember Learning Module
Set the number of reversible module iterations K.
Set $\mathbf{M}_0$ to sample from the endmember distribution.
Initialize $\mathbf{M}_0 = \mathbf{z}$ and Jacobian determinant $logdet = 0$.
$While\ k < K$
normalization : $\mathbf{M}_k^1 = a_k \cdot \mathbf{M}_{k-1} + b_k$,
$logdet_k^1 = logdet_{k-1} + log\left|\frac{1}{a_k}\right|$
permutation operation : $\mathbf{M}_k^2 = conv(\mathbf{M}_k^1) = \mathbf{W}\mathbf{M}_k^1$,
$logdet_k^2 = logdet_k^1 + log\left|\mathbf{W}^{-1}\right|$
affine coupled transformation :
$\mathbf{M}_{k,1}^2, \mathbf{M}_{k,2}^2 = split(\mathbf{M}_k^2)$
$\mathbf{h}_k^1 = sigmoid(conv(sigmoid(conv(\mathbf{M}_{k,1}^2))))$
$\mathbf{h}_k^2 = sigmoid(conv(sigmoid(conv(\mathbf{X}))))$
$\mathbf{h}_k = (\mathbf{h}_k^1, \mathbf{h}_k^2)$
$\mathbf{f}_k, \mathbf{g}_k = split(\mathbf{h}_k)$
$\mathbf{M}_{k,2}^2 = \mathbf{f}_k(\mathbf{M}_{k,2}^2 + \mathbf{g}_k)$
$\mathbf{M}_k = concat(\mathbf{M}_{k,1}^2, \mathbf{M}_{k,2}^2)$
$logdet_k = logdet_k^2 + log\left|\mathbf{f}_k^{-1}\right|$
k=k+1
Abundance Guidance Module
$feature_1 = ReLU(BN(conv(\mathbf{X})))$
$feature_2 = ReLU(BN(conv(feature_1)))$
$\mathbf{A} = Softmax(conv(feature_2))$
Hyperspectral Image Reconstruction
$\hat{\mathbf{X}} = \mathbf{M}_K \cdot \mathbf{A}$
Return the vector sample from the endmember distribution M_K, abundance maps A, reconstructed image $\hat{X}$, and logarithm Jacobian determinant $logdet_K$

C. Endmember Learning Module

The input of the network are randomly sampled from a simple distribution, and we use the standard normal distribution in this article. Aided by hyperspectral images, the input is transformed into the endmember vector with spectral variability after a series of linear transformations. Meanwhile, we can obtain the corresponding function transformation and Jacobian determinant which represents the scaling factor from

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the standard normal distribution to the endmember distribution. The mathematical expression between the endmember distribution and the standard normal distribution is

$$p(\mathbf{y}) = \pi(\mathbf{z}) \cdot |\det(\mathbf{J}^{-1})| \tag{5}$$

where $\mathbf{z} \in R^{L\times P}$ follows a simple distribution $\pi(\mathbf{z})$, and $\mathbf{y} \in R^{L\times P}$ follows the endmember distribution $p(\mathbf{y})$. $\mathbf{J} \in R^{L\times L}$ denotes the Jacobian matrix of the transformation from $\mathbf{z}$ to $\mathbf{y}$, and $|\det(\cdot)|$ represents the absolute value of determinant, which is a scalar. Given that $\pi(\mathbf{z})$ is known, we need to compute $|\det(\mathbf{J}^{-1})|$ to get the mathematical expression for $p(\mathbf{y})$. This module consists of three components: normalization, permutation operation, and affine coupled transformation.

1) Normalization: In order to prevent gradient explosion or disappearance and to make the training process more stable, we set up a normalization layer. The standardization module normalizes the input data on the channel dimension. This can be written as a linear transformation, i.e.,

$$\mathbf{M}_1 = a\mathbf{M}_0 + b \tag{6}$$

$$\mathbf{M}_0 = (\mathbf{M}_1 - b)/a \tag{7}$$

$$\left|\det\left(\mathbf{J}_{\text{norm}}^{-1}\right)\right| = \frac{1}{a} \tag{8}$$

where $\mathbf{M}_0$ and $\mathbf{M}_1$ denotes the input and output of normalization layer.

2) Permutation Operation: In order to prevent the generated data from completely copying parts of the original image, we use permutation transformation to imitate the flow model [43]. There are three methods.

1) Flip in the channel dimension.
2) Randomly disrupt the sequence of channels.
3) Use 1×1 convolution to scramble.

Because 1×1 Convolution can achieve the purpose of disrupting channels and is easy to calculate, we used 1×1 convolution in our model [43] as

$$\mathbf{M}_2 = \text{conv}(\mathbf{M}_1). \tag{9}$$

In this process, the dimensions of input and output are the same, and the dimensions are both $L \times P$, where L is the number of image channels and P is the number of endmembers. The number of convolution kernel for a 1×1 convolution is $L \times P$. The weight $\mathbf{W}$ of the 1×1 convolution is randomly initialized. In this case, the Jacobian matrix is also easy to calculate. Considering the complexity of weight $\mathbf{W}$, we can use LU decomposition to calculate its determinant. We can write this in a mathematical form

$$\mathbf{M}_2 = \mathbf{W}\mathbf{M}_1 \tag{10}$$

$$\mathbf{M}_1 = \mathbf{W}^{-1}\mathbf{M}_2 \tag{11}$$

$$\left|\det\left(\mathbf{J}_{\text{per}}^{-1}\right)\right| = |\mathbf{W}^{-1}| \tag{12}$$

where $\mathbf{M}_1$ and $\mathbf{M}_2$ denote the input and output of permutation operation.

3) Affine Coupled Transformation: Influenced by normalizing flow [43], the majority of the proposed endmember learning module uses affine coupled transformation, which makes the Jacobian determinant easier to compute. The module is divided into two steps. First, we obtain the scale vector and the bias vector by applying a split transformation to input $\mathbf{M}_2$. Then, the input vector is transformed in a linear way. The process is as follows, which is shown in Fig. 2.

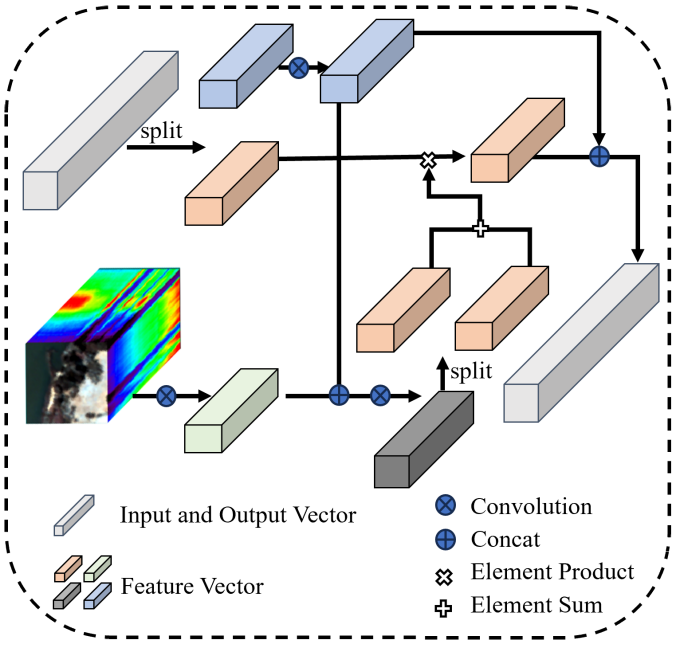


Fig. 2. Affine-coupled transformation in endmember learning module. Different color cubes represent feature vectors of different dimensions.

The input vector $\mathbf{M}_2 = [x_1, \ldots, x_L]$ is divided into two parts, which are $\mathbf{x}^1 = [x_1, \ldots, x_m]$ and $\mathbf{x}^2 = [x_{m+1}, \ldots, x_L]$. The upper vector $\mathbf{x}^1$ and hyperspectral image are input into two networks and we obtain the output scale vector $\mathbf{f} = [f_{m+1}, \ldots, f_L]$ and bias vector $\mathbf{g} = [g_{m+1}, \ldots, g_L]$ through some transformation. The two vectors $\mathbf{f}$ and $\mathbf{g}$ obtained have the same dimension as $\mathbf{x}^2$. To be specific, $\mathbf{x}^1$ and hyperspectral image are the input of two convolutional modules, in which each module is composed two layers. Two layers are both composed of a 3×3 convolution and the sigmoid activation function. The two output vectors obtained by two convolutional modules are spliced. The concatenated vector is shuffled with a convolutional layer and then divided into two parts, a scale and a bias vector. The mathematical form is shown as

$$\mathbf{h}_1 = \text{sigmoid}\left(\text{conv}\left(\text{sigmoid}\left(\text{conv}\left(\mathbf{x}^1\right)\right)\right)\right) \tag{13}$$

$$\mathbf{h}_2 = \text{sigmoid}(\text{conv}(\text{sigmoid}(\text{conv}(\mathbf{X})))) \tag{14}$$

$$\mathbf{f}, \mathbf{g} = \text{split}(\text{conv}(\mathbf{h}_1, \mathbf{h}_2)) \tag{15}$$

where $\mathbf{x}^1$ denotes the upper vector of affine coupled transformation and $\mathbf{X}$ is hyperspectral image.

The upper vector $\mathbf{x}^1$ is the upper half of the output vector after identity transformation, which is called $\mathbf{y}^1 = [y_1, \ldots, y_m]$. The lower half of the input vector $\mathbf{x}^2$ is added to the bias vector $\mathbf{g}$, and multiplied with scale vector $\mathbf{f}$ at the element level to get the lower half of the new vector $\mathbf{y}^2 = [y_{m+1}, \ldots, y_L]$. Merge $\mathbf{y}^1$ and $\mathbf{y}^2$ to get the new vector $\mathbf{y}$. The mathematical form is shown as

$$y_i = x_i, \quad i = 1, \ldots, m \tag{16}$$

$$y_i = f_i(x_i + g_i), \quad i = m+1, \ldots, L. \tag{17}$$

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The inverse form of affine coupled transformation is

$$x_i = y_i, \quad i = 1, \ldots, m \tag{18}$$

$$x_i = y_i / f_i - g_i, \quad i = m+1, \ldots, L. \tag{19}$$

In this case, the transformation of x to y is a linear mapping, and we can compute the Jacobian matrix of it as

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{pmatrix} 1 & & & 0 & 0 & 0 \\ & \ddots & & 0 & 0 & 0 \\ & & 1 & 0 & 0 & 0 \\ * & * & * & f_{m+1} & & \\ * & * & * & & \ddots & \\ * & * & * & & & f_L \end{pmatrix} \tag{20}$$

where $*$ is unknown value which do not affect the calculation of the matrix determinant because of the zero matrix in the top right corner and the matrix determinant is

$$\left|\det\left(\mathbf{J}_{\text{affine}}^{-1}\right)\right| = \left(\det\left|\frac{\partial \mathbf{y}}{\partial \mathbf{x}}\right|\right)^{-1} = \left(\prod_{i=m+1}^{L} f_i\right)^{-1}. \tag{21}$$

We have introduced the specific structure of the three components and given the Jacobian determinant of each step. The expression of random variables following the endmember distribution will be described in the next paragraphs.

For each vector $\mathbf{z}_0 \sim N(0, 1)$ randomly sampled, the vector $\mathbf{z}_1$ is obtained by the invertible module in the first iteration, which can be obtained by transformation formula

$$p(\mathbf{z}_1) = \pi(\mathbf{z}_0) \cdot \left|\det\left(\mathbf{J}_{(1)}^{-1}\right)\right| \tag{22}$$

where $\mathbf{J}_{(1)}$ denotes the Jacobian matrix of transformation from $\mathbf{z}_0$ to $\mathbf{z}_1$, i.e.,

$$\mathbf{J}_{(1)} = \frac{\partial \mathbf{z}_1}{\partial \mathbf{z}_0} = \begin{bmatrix} \frac{\partial z_{11}}{\partial z_{01}} & \cdots & \frac{\partial z_{11}}{\partial z_{0L}} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_{1L}}{\partial z_{01}} & \cdots & \frac{\partial z_{1L}}{\partial z_{0L}} \end{bmatrix} \tag{23}$$

$$\left|\det\left(\mathbf{J}_{(1)}^{-1}\right)\right| = \left|\det\left(\mathbf{J}_{\text{norm}}^{-1}\right)\right| \cdot \left|\det\left(\mathbf{J}_{\text{per}}^{-1}\right)\right| \cdot \left|\det\left(\mathbf{J}_{\text{affine}}^{-1}\right)\right|. \tag{24}$$

The vector $\mathbf{z}_2$ is obtained by the second invertible module according to the chain rule and we can get the result

$$p(\mathbf{z}_2) = \pi(\mathbf{z}_0) \cdot \left|\det\left(\mathbf{J}_{(1)}^{-1}\right)\right| \cdot \left|\det\left(\mathbf{J}_{(2)}^{-1}\right)\right|. \tag{25}$$

The endmember vector $\mathbf{y}$ is obtained after K identical multiple transformations, each of which contains the three transformations mentioned earlier, and its probability is calculated as

$$p(\mathbf{y}) = \pi(\mathbf{z}) \cdot \left|\det\left(\mathbf{J}_{(1)}^{-1}\right)\right| \cdot \left|\det\left(\mathbf{J}_{(2)}^{-1}\right)\right| \cdot \cdots \cdot \left|\det\left(\mathbf{J}_{(K)}^{-1}\right)\right|. \tag{26}$$

Since $\mathbf{z}$ follows a simple distribution, we usually use Gaussian distribution and the probability density function is:

$$\pi(\mathbf{z}) = \frac{1}{\left(\sqrt{2\pi}\right)^L} e^{-\frac{1}{2}||\mathbf{z}||^2}. \tag{27}$$

The probability density function of endmember distribution is

$$p(\mathbf{y}) = \frac{1}{\left(\sqrt{2\pi}\right)^L} e^{-\frac{1}{2}||\mathbf{y}||^2} \cdot \prod_{k=1}^{K} \left|\det\left(\mathbf{J}_{(k)}^{-1}\right)\right| \tag{28}$$

where $\prod_{k=1}^{K} |\det(\mathbf{J}_{(k)}^{-1})|$ can be estimated by the module and the proof is shown in Section II-E.

D. Abundance Guidance Module

In the abundance guidance module, we adopt the encoder structure and obtain abundance maps inspired by the unmixing algorithm of convolutional autoencoder. This module is designed by using the convolutional neural networks. The first layer is the input layer, which takes the hyperspectral image as the input of the module. 3×3 convolution is used to reduce the channel dimension from L to $L//2$ in the second layer and the third layer is 3×3 convolution with $L//4$ feature map, where // indicates exact division. We add batch normalization after second and third convolution layers to speed up network training, and the nonlinear activation function is set to ReLU. Finally, the vector dimension is reduced to the number of endmembers P in the image after 3×3 convolution. Since the convolution kernel we use is able to obtain information about the surrounding adjacent pixels, we are able to obtain a certain amount of spatial information when learning the abundance map. The abundance maps satisfy ASC and ANC through Softmax activation function. It can be formulated as follows:

$$\begin{aligned} \mathbf{A} = \text{Softmax}(\text{conv}(\text{ReLU}(BN(\text{conv} \\ \times(\text{ReLU}(BN(\text{conv}(\mathbf{X})))))))) \end{aligned} \tag{29}$$

where $\mathbf{X}$ denotes hyperspectral image and $\mathbf{A}$ denotes the abundance map.

Similar to autoencoder unmixing, the feature map obtained by this module is considered to be abundance. According to the LMM considering spectral variability, we multiply the estimated abundance maps linearly with the endmembers obtained by the endmember learning module pixel by pixel to obtain the reconstructed image and calculate the reconstruction loss. By minimizing the reconstruction loss, the distribution obtained by the endmember learning module is closer to the unknown distribution. In this way, the module obtains abundance and guides the learning of endmember distribution.

E. Theoretical Analysis of Reversibility

If endmember learning module is invertible, we can guarantee that every step of the change experienced in our model is known. One-to-one correspondence means that different noises will correspond to different endmembers, which mitigates the problem of model collapse. Controllable makes the process of endmember generation be stable and hard to mutation. In order to guarantee these properties, we need to prove the reversibility.

Theorem 1 (Reversibility): For a simple prior distribution $p_z(\mathbf{z})$ and a complex distribution $p_x(\mathbf{x})$, suppose a mapping relationship between $p_z(\mathbf{z})$ and $p_x(\mathbf{x})$ is established through a

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series of invertible transformations. We define $\mathbf{x} = f(\mathbf{z})$. If f satisfies (i) f is invertible and (ii) the Jacobian determinant is computable, then

$$p_x(\mathbf{x}) = p_z(\mathbf{z})|\det \mathbf{J}_{f^{-1}}(\mathbf{x})| = p_z(\mathbf{z})|\det \mathbf{J}_f(\mathbf{z})|^{-1}.$$

Proof: Suppose $\mathbf{z}_0 \sim p_z(\mathbf{z})$, $\mathbf{x} \sim p_x(\mathbf{x})$, and $\mathbf{z}_1 = f_0(\mathbf{z}_0), \mathbf{z}_2 = f_1(\mathbf{z}_1), \ldots, \mathbf{x} = f_{k-1}(\mathbf{z}_{k-1})$ are invertible transformations. Let $\mathbf{x} = f(\mathbf{z}) = f_{k-1} \circ \ldots \circ f_1 \circ f_0(\mathbf{z})$, where f is continuous and invertible.

$\int_z p_z(\mathbf{z})dz = 1 = \int_x p_x(\mathbf{x})dx$ and if A and B are viewed as uniformly distributed over a minimal region, the integrals over that interval are equal, i.e., $p_z(\mathbf{z})dz = p_x(\mathbf{x})dx$

$$p_x(\mathbf{x}) = p_z(\mathbf{z})\frac{d\mathbf{z}}{d\mathbf{x}} \tag{30}$$

(we can apply Radon–Nikodym theorem to prove it).

Due to $\mathbf{x} = f(\mathbf{z})$ and f is invertible, then

$$p_x(\mathbf{x}) = p_z(\mathbf{z})\left|\frac{\partial f^{-1}(\mathbf{x})}{\partial \mathbf{x}}\right| \tag{31}$$

where $\mathbf{z} = f^{-1}(\mathbf{x})$ and we take the absolute value of $(\partial f^{-1}(\mathbf{x})/\partial \mathbf{x})$ because it may be negative.

For multivariate distribution, $(\partial f^{-1}(\mathbf{x})/\partial \mathbf{x})$ is Jacobian determinant, i.e., $det(\mathbf{J}_{f^{-1}}(\mathbf{x}))$. With $|\det((\partial f^{-1}(\mathbf{x})/\partial \mathbf{x}))| = |\det((\partial f(\mathbf{z})/\partial \mathbf{z}))|^{-1}$, we find

$$p_x(\mathbf{x}) = p_z(\mathbf{z})|\det \mathbf{J}_{f^{-1}}(\mathbf{x})| = p_z(\mathbf{z})|\det \mathbf{J}_f(\mathbf{z})|^{-1} \tag{32}$$

where we finish proof the theorem 1.

In Section II-C, we provide the inverse transformation form and Jacobian determinant of the three transformations, respectively. Invertibility of every transformation means f is a composition of several invertible functions and therefore also invertible, which satisfies Theorem 1 (i). By the chain rule, the Jacobian determinant of the transformation from standard normal distribution to endmember distribution is the product of multiple Jacobian determinants which satisfies Theorem 1 (ii).

Therefore, endmember learning module is reversible, i.e.,

$$p(\mathbf{y}) = \pi(\mathbf{z}_0)\left|\det\left(\mathbf{J}_f^{-1}\right)\right| \tag{33}$$

$$\pi(\mathbf{z}_0) = p(\mathbf{y})\left|\det\left(\mathbf{J}_f\right)\right| \tag{34}$$

where f includes multiple transformation and $\mathbf{y} = f(\mathbf{z}_0)$.

F. Loss Function

The overall loss in our proposed method consists of four parts, which is formulated as

$$L_{\text{total}} = L_{\text{rec}} + \alpha L_{\text{sparse}} + \beta L_{\text{edm}} + \gamma L_{\text{det}}. \tag{35}$$

L_{rec} denotes reconstruction loss. As a commonly used function in deep learning unmixing, spectral angle distance (SAD) shows excellent performance. The first term L_{rec} is defined as

$$L_{\text{rec}} = \frac{1}{N}\sum_{n=1}^{N}\arccos\left(\frac{<\mathbf{x}_n, \hat{\mathbf{x}}_n>}{||\mathbf{x}_n|| \cdot ||\hat{\mathbf{x}}_n||}\right). \tag{36}$$

Since abundance maps are sparse, we need to use regularization terms. We choose the $L_{(1/2)}$ norm, and then L_{sparse} is

$$L_{\text{sparse}} = \left||\mathbf{A}|\right|_{\frac{1}{2}} = \sum_{n=1}^{N}\sum_{p=1}^{P}|a_{np}|^{\frac{1}{2}}. \tag{37}$$

For the third term, we give the constraint of the endmember. We divide the training process into two stages. In order to control the uncertainty of the model, we provide a good initial direction at the beginning of the model training. We realize this idea by using the endmembers extracted from vertex component analysis (VCA). In this case, the endmember loss is the MSE between the endmember obtained by distribution and the vector extracted by VCA, i.e.,

$$L_{\text{edm}} = \frac{1}{N}\left(\sum_{p=1}^{P}\sum_{n=1}^{N}\left|\left|\tilde{\mathbf{M}}_{np} - \mathbf{M}_p^{\text{VCA}}\right|\right|_2^2\right)^{\frac{1}{2}} \tag{38}$$

where $\tilde{\mathbf{M}}_{np}$ is the pth estimated endmember corresponding to the nth pixel, and $\mathbf{M}_p^{\text{VCA}}$ is the pth endmember obtained by the VCA algorithm. Considering the magnitude of this loss, we add the root operation when conducting experiments.

In the second stage, in order to prevent the variance of the endmember distribution from being too large, we want to obtain a relatively small MSE between the endmember and its mean, i.e.,

$$L_{\text{edm}} = \frac{1}{NP}\sum_{p=1}^{P}\sum_{n=1}^{N}\left|\left|\tilde{\mathbf{M}}_{np} - \frac{1}{N}\left(\sum_{n=1}^{N}\tilde{\mathbf{M}}_{np}\right)\right|\right|_2^2. \tag{39}$$

Since we give an explicit expression for the endmember distribution, i.e.,

$$p_X(\mathbf{x}) = p_Z(\mathbf{z})\prod_{k=1}^{K}\left|\det \mathbf{J}_{f_k}.(\mathbf{z}_{k-1})\right|^{-1}. \tag{40}$$

According to the idea of statistics, after bringing the sample into the distribution function, the function can also be regarded as the likelihood function. The idea of maximum likelihood estimation is finding the parameter that can make the existing sample appear with maximum probability. That is, we will optimize the entire network by maximizing (40) or its logarithm in the following equation, i.e.,

$$\log p_X(\mathbf{x}) = \log p_Z(\mathbf{z}) - \sum_{k=1}^{K}\log|\det \mathbf{J}_{f_k}(\mathbf{z}_{k-1})|. \tag{41}$$

The output of a reversible neural network contains the Jacobian determinant of the map, and both $p_Z(\mathbf{z})$ and $\det \mathbf{J}_{f_k}(\mathbf{z}_{k-1})$ in (41) can be obtained. Since the updating of the neural network is usually to minimize the loss function, the negative value of (41) is chosen for the fourth part of the loss function. Therefore, the fourth item L_{det} can be defined as

$$L_{\text{det}} = \exp\left\{-\log p_Z(\mathbf{z}) + \sum_{k=1}^{K}\log|\det \mathbf{J}_{f_k}(\mathbf{z}_{k-1})|\right\}. \tag{42}$$

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III. Experiments

In this section, we use both real and synthetic datasets to evaluate the performance of our proposed reversible generative network (Rev-Net) method, which we compare with some advanced hyperspectral methods, including VCA [44] and fully constrained least squares (FCLSU) [45], an NMF-based method as robust nonnegative matrix factorization (rNMF) [46], convolutional neural network autoencoder [convolutional autoencoder for spectral spatial hyperspectral unmixing (CNNAEU)] [47], extended linear mixing model (ELMM) [33], perturbed linear mixing model (PLMM) [34], generalized linear mixing model (GLMM) [48], as well as generative model unmixing method as deep generative endmember model (DeepGUn) [37], deep generative model for spatial–spectral unmixing (DGMSSU) [49], and VAE [40].

A. Experimental Setup

1) Evaluation Metrics: To quantitatively evaluate the performance of our proposed unmixing method, we calculate the accuracy of the obtained endmembers and abundance maps, respectively. For the endmembers, we use endmember spectral angular distance (eSAD) to measure the difference between the obtained endmembers and the reference endmembers. For abundance maps, we use the mean square error (MSE) to measure the difference between the estimated abundance and the reference abundance

$$\text{eSAD} = \frac{1}{B}\sum_{i=1}^{B}\arccos\left(\frac{\mathbf{m}_i^T\hat{\mathbf{m}}_i}{\|\mathbf{m}_i\|_2\|\hat{\mathbf{m}}_i\|_2}\right) \tag{43}$$

$$\text{MSE} = \frac{1}{N}\sum_{n=1}^{N}\left|\left|\mathbf{h}_n - \hat{\mathbf{h}}_n\right|\right|^2 \tag{44}$$

where $\mathbf{m}_i$ denotes the ith reference endmember and $\hat{\mathbf{m}}_i$ is the ith estimated endmember. $\mathbf{h}_i$ denotes the reference abundance of the ith pixel, and $\hat{\mathbf{h}}_i$ is the estimated abundance of the ith pixel. The smaller the eSAD and MSE, the closer the obtained endmembers and abundance maps are to the reference value, which means the unmixing performance is better.

2) Hyperparameter Settings: For hyperparameter used in the Rev-Net, as described in Section II-F, the training process is divided into two stages where the endmember loss changes, and the hyperparameters in (33) are different. In the first stage, the $\alpha = 1e-2$, $\beta = 100$, $\gamma = 1e-10$ and in the second stage, $\alpha = 1e-2$, $\beta = 10$, $\gamma = 1$. In addition, the optimizer is set to Adam optimizer and the learning rate is set to 3e-4. To be fair, the endmember initialization of all methods takes into account the results of VCA extraction, and the abundance maps of ELMM, PLMM, GLMM, and DeepGUn are initialized with the abundance maps obtained by FCLSU. The endmember number P of hyperspectral image is obtained based on HySime algorithm before model training [50].

In order to check the sensitivity of the parameters and the convergence of the algorithm, we do some experiments. As for the sensitivity of the parameter, we examine the influence of the value of K in the endmember learning module on the experimental results. We set K = [1,5,10,20,50,70,100], respectively, to conduct experiments on synthetic datasets. The experimental results are shown in the Fig. 3. We can find that the value of K greater than ten has little influence on the experimental results. In the actual experiment, considering the stability and complexity of the model, we finally chose $K = 50$. Regarding the convergence of the algorithm, we provide the MSE between the resulting abundance and the true abundance change as the number of training increases, which is shown in the Fig. 4, so we believe that our algorithm is convergent.

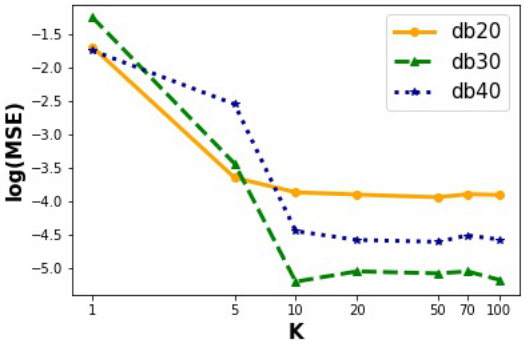


Fig. 3. Influence of the K value on the experimental results under different SNR. The abscissa is the value of K, and the ordinate is the logarithm of MSE between the estimated abundance and the true abundance.

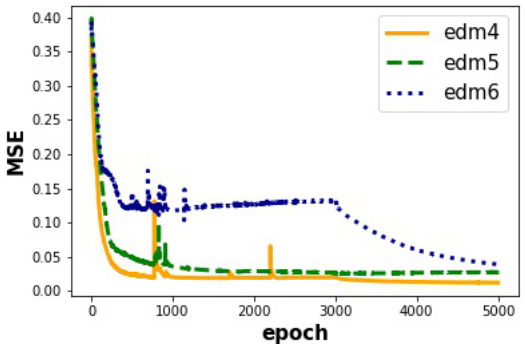


Fig. 4. MSE between the estimated abundance and the true abundance as the epoch increases in training process in the case of 4,5, and 6 endnumber.

B. Data Description

1) Real Data:

1) The Samson dataset is a real dataset collected by the SAMSON sensor, which is often used for hyperspectral unmixing. The original dataset size is 952 × 952, and each pixel is recorded on 156 channels with a wavelength of 401-889 nm. Due to limitations such as large datasets and expensive calculations, we usually use adapted images containing 95 × 95 pixels. There are three main materials in the image, that is, soil, water, and tree.
2) The Cuprite dataset is a real dataset collected by Aviris, which is often used for hyperspectral unmixing. The original dataset size is 250 × 190, and each pixel is recorded on 224 channels with a wavelength of 370-2480 nm. Since some of the channels are noise and water absorption channels, we used data from 188 channels in the experiment.

2) Synthetic Data:

1) *Generating Endmember With Variability:* We used the United States Geological Survey (USGS) spectral library

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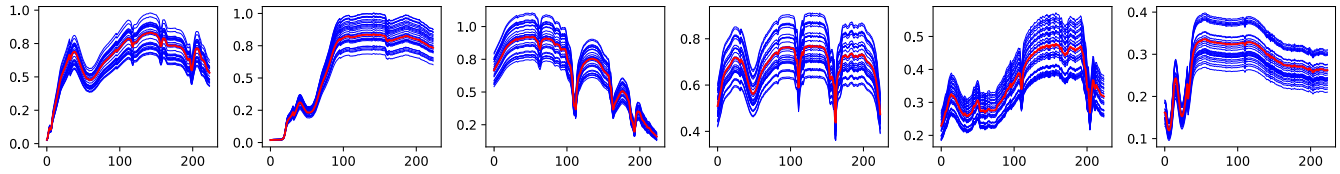


Fig. 5. Reference endmembers (red line) and 30 corresponding endmembers that increase spectral variability.

TABLE I

eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SAMSON DATASET (10^{-2})

Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
eSAD	6.67	28.19±0.23	7.69±1.19	45.03	12.38	6.60	43.88	13.23	5.62±0.18	**3.92**±0.38
MSE	22.07	55.12±0.33	6.56±3.31	21.40	23.69	22.07	30.59	17.65	14.8±0.68	**5.82**±2.26

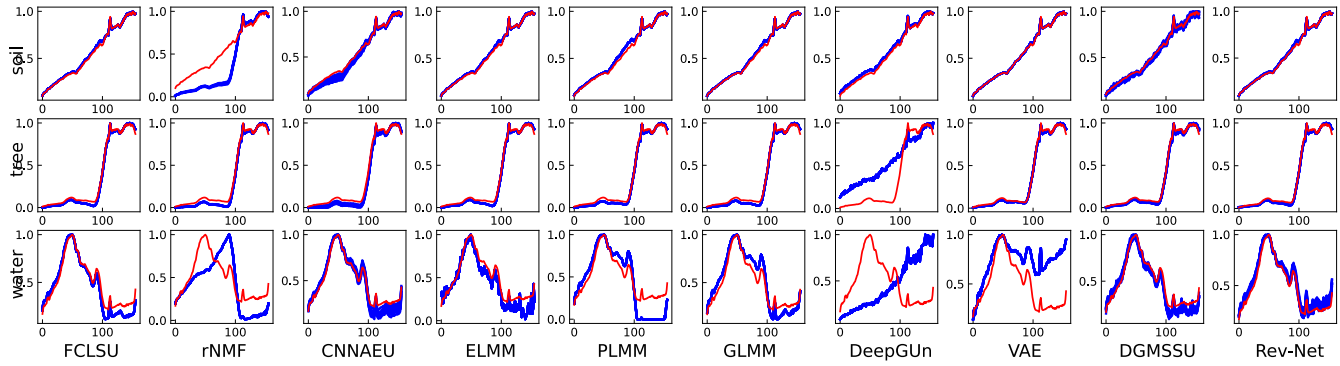


Fig. 6. Soil, water, and tree endmembers extracted from the Samson dataset using ten methods, which run ten times. The red curve in the figure represents the true endmembers of Samson dataset, and the blue curve represents the endmembers estimated by ten algorithms at each time.

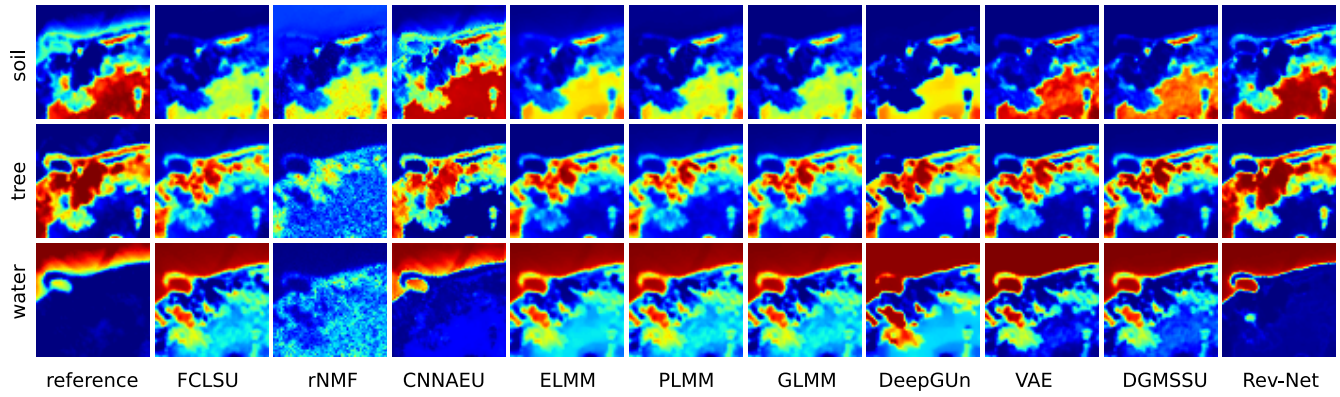


Fig. 7. Estimated abundance maps for Samson dataset. Abundance values represented by colors range from blue (0) to red (1).

TABLE II

eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SYNTHETIC SAMSON (10^{-2})

endmember	Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
1st	eSAD	2.07	34.18±0.75	4.49±1.96	18.76	3.53	2.19	3.10	**1.02**	2.60±0.28	1.73±0.27
	MSE	6.58	17.06±0.12	2.38±1.14	5.06	6.58	6.57	8.39	3.38	3.56±0.24	**2.32**±1.00
2nd	eSAD	4.95	7.02±0.16	8.23±1.10	22.54	5.01	5.19	41.64	**3.33**	3.67±0.15	3.66±0.16
	MSE	3.31	15.44±0.11	2.20±1.19	4.24	3.97	3.30	4.58	3.92	2.84±0.21	**0.38**±0.11
3rd	eSAD	12.99	43.38±0.11	10.34±2.76	93.78	28.61	12.41	86.89	35.36	10.61±0.34	**6.36**±1.05
	MSE	12.17	22.62±0.12	**1.98**±1.30	12.10	13.13	12.20	17.61	10.34	8.42±0.44	3.13±1.25

which is captured with the Beckman5270 spectrometer [51], from which we select the spectra of a number of materials as the reference endmembers, and the spectra all included 224 continuous bands ranging from 400 to

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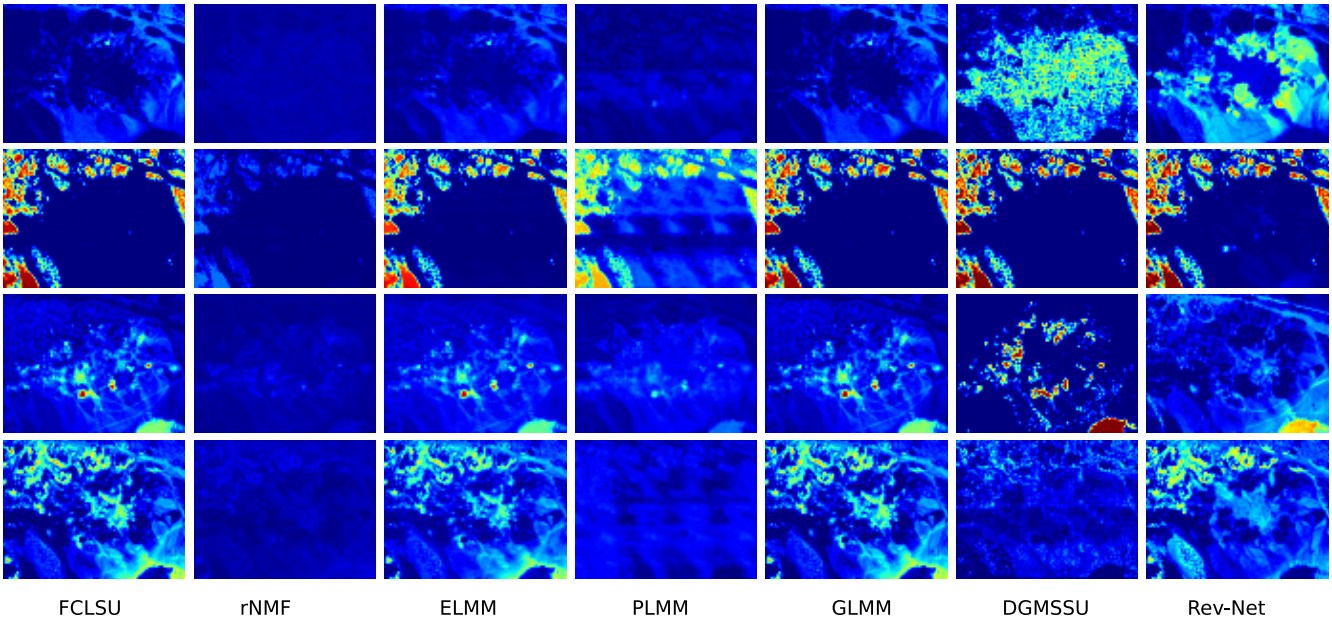


Fig. 8. Estimated abundance maps for Cuprite dataset. Abundance values represented by colors range from blue (0) to red (1).

2500 nm. To add spectral variability, we use an endmember scaling factor. In the specific generation process, in order to make the endmember at each pixel have variability, we multiply the endmember of each pixel by a randomly generated scale factor, where the scale factor of the endmember of each pixel is the same for all bands, and our scale factor follows a uniform distribution of [0.8,1.2] [35]. Fig. 5 shows the schematic after we added the spectral variability for the reference spectrum, where the red line is the reference endmember and the blue line are the transformed endmembers.

2) *Generating Synthetic Abundance Data:* We generated some abundance maps of 100 × 100 pixels with spatial information [52]. In the generation process, we split the image into 10 × 10 square regions, each containing a pure spectrum, mix the pixels using a spatial filter, and control the abundance maximum value to 0.9. Abundance matrix is guaranteed both ANC and ASC.

3) *Generating the Mixed Pixels:* The generation of each simulated hyperspectral data is generated by (4), and then the additive noise of the fixed signal-to-noise ratio (SNR) is introduced. In our experiment, Gaussian noise is added. Data1 consist of three datasets, whose endmember number is set to three and SNR is set to 20, 30, and 40db, respectively, so as to explore the effectiveness of our proposed unmixing method on datasets with different SNR. Data2 consist of three datasets, all of which make SNR set to 30 db and endmember number set to 4, 5, and 6 to explore the effectiveness of our proposed unmixing method on datasets with different endmember number. Data3 consist of three datasets, all of which make SNR, respectively, set to 30 db and endmember number set to 4, 5, and 6 to explore the effectiveness of our proposed unmixing method on datasets without spectral variability.

C. Experimental Result

1) Real Data: We use the table to show the comparison of unmixing results using deep learning unmixing methods and classical methods considering spectral variability with Rev-Net method. We use (43) and (44) provided above to calculate the difference between the estimated endmembers as well as the abundance matrix and the reference value, which is regarded as the basis to judge the unmixing performance. To add persuasiveness, in the table, we show the mean values of eSAD and MSE for running ten experiments and use bolded best results. For the methods that need to be initialized, we use the results of the VCA and FCLSU shown in the table as initialization. The table shows the mean and standard deviation of the results of the ten runs, and the results retain two significant digits. For algorithms with standard deviation less than $1e-5$, such as ELMM, PLMM, and GLMM, we only show the mean value and do not show the standard deviation in the form of 0 in the table.

Table I shows the unmixing effect of the ten methods on the real dataset Samson. We can see that the mean value of eSAD between the estimated endmember of Rev-Net and the reference is 3.92×10^{-2}. The MSE between the estimated abundance matrix and the true value is 5.82×10^{-2}. The minimum values of the two indexes of the other nine methods are 5.62×10^{-2} and 6.56×10^{-2}, respectively. From the quantitative indicators, we can find that our method has obvious advantages compared with other methods. In order to more intuitively understand the experimental results, we provide the visualized results of endmembers and abundance.

Fig. 6 shows the endmembers extracted by ten methods from Samson dataset and the reference endmembers. The blue curve represents the endmembers obtained by the various unmixing methods, each of which is run ten times, and the red curve represents the reference endmembers. From Fig. 6,

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TABLE III
eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SYNTHETIC DATA1 (10^{-2})

data	Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
db20	eSAD	5.61	7.32±0.22	13.04±3.28	5.49	12.07	**4.81**	30.15	6.48	7.79±0.19	5.57±0.06
	MSE	4.34	21.49±0.27	14.37±7.83	2.80	20.28	4.27	89.22	3.51	2.92±0.20	**1.82**±0.18
db30	eSAD	6.35	12.33±0.79	14.39±3.00	6.08	6.17	6.74	25.77	9.56	8.75±0.20	**5.91**±0.03
	MSE	7.63	20.35±1.21	9.22±1.86	2.86	6.12	7.60	8.51	6.75	3.72±0.20	**0.57**±0.01
db40	eSAD	3.25	7.66±0.08	15.48±8.71	3.25	3.23	**3.11**	26.14	6.36	6.02±0.18	3.35±0.21
	MSE	13.45	23.18±0.23	12.46±6.31	9.65	5.66	13.50	73.60	10.98	9.88±1.36	**1.05**±0.14

TABLE IV
eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SYNTHETIC DATA1-SNR = 30 DB (10^{-2})

endmember	Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
1st	eSAD	0.91	4.10±0.78	4.56±1.22	0.97	**0.79**	2.56	1.77	2.25	2.11±0.17	0.86±0.01
	MSE	3.79	5.54±0.18	2.56±0.89	1.10	1.95	3.81	4.05	2.81	1.63±0.12	**0.14**±0.01
2nd	eSAD	0.86	3.40±0.41	10.67±3.75	1.03	1.96	2.23	30.96	2.35	4.25±0.76	**0.83**±0.01
	MSE	2.67	1.85±0.07	2.11±0.61	0.69	2.33	2.63	3.15	0.65	1.10±0.09	**0.13**±0.01
3rd	eSAD	17.27	29.47±2.75	27.94±7.40	16.25	15.77	**15.44**	44.58	24.09	19.90±0.58	16.06±0.12
	MSE	1.18	12.96±1.06	4.54±0.95	1.08	1.84	1.16	1.31	3.30	0.99±0.08	**0.30**±0.01

we can find that Rev-Net is closer to the true value than other methods. All methods except rNMF are relatively accurate in the acquisition of soil endmember. Most of the methods have some slight deviations from the reference values, but the endmember curves estimated by VAE and Rev-Net almost coincide with the red curve. In the acquisition of water endmember, other methods have obvious deviations, but the reference endmembers within the range of the ten endmembers extracted by Rev-Net. Although there is variance between the results of ten runs, Rev-Net are obviously better than other methods.

Fig. 7 shows the abundance map of the Samson dataset obtained by various methods. In the abundance maps, we can find that Rev-Net has advantage in the acquisition of abundance corresponding to the endmembers of soil and tree. In the abundance map corresponding to the soil, the color in the lower part of the image is purer on the basis of restoring the basic shape of the map. For the abundance map of tree, the results of our estimation and CNNAEU are closer to the reference abundance map.

In order to evaluate each endmember and the corresponding abundance estimated by ten methods more clearly and accurately, we calculated the eSAD between the estimated value of each endmember estimated by each method and the reference, as well as the MSE between the corresponding abundance matrix and the reference. Table II shows the eSAD and MSE of the corresponding abundance of each endmember in the Samson dataset, where we can see the same analysis as our visualization results above.

In addition, the Cuprite dataset contains multiple endmembers, and the corresponding abundances of some endmembers are shown in Fig. 8. We can find that Rev-Net achieves better unmixing effect compared with other methods.

2) Synthetic Data: For synthetic experiments, to explore the feasibility of our approach when the SNR changes or the number of endmembers changes, we generate two datasets. The Data1 dataset consists of three datasets with SNR sets of 20, 30, and 40db, respectively. In Table III, we show the average eSAD and MSE of the endmembers estimated by ten methods under each SNR. We can see that Rev-Net obtains higher accuracy under different SNRs. For the acquisition of endmembers, Rev-Net is also competitive because the accuracy of the endmember estimated by Rev-Net is always better. On the whole, Rev-Net achieves better unmixing effect under different SNRs.

In order to understand the unmixing effect in more detail, we also provide the accuracy of the three endmembers corresponding to the abundance estimation under 30 db SNR. Table IV shows the average eSAD and MSE for the estimator of each endmember and with reference values. We can find that the smallest eSAD for estimation of the first and third endmembers are PLMM and GLMM, respectively, and the second endmember estimated by Rev-Net reaches the minimum. In the endmember estimation that does not achieve the optimal result, the difference between unmixing results obtained by Rev-Net and the optimal value is little. The average MSE of Rev-Net reaches the minimum value and is much smaller than that of other methods.

In order to make the unmixing performance of the ten methods on Data1 more intuitive, we show the spectral curves and abundance maps of the estimated endmembers and reference endmembers of each method. Fig. 9 shows the endmember estimation of Data1 -30 db. Fig. 10 shows the abundance maps estimated by various methods at 30 db. From the figure, we can find that most methods can give relatively accurate endmember and abundance estimation, while the abundance obtained by Rev-Net is smoother at the edge of the graph and purer in the interior color of the graph, which is closer to the reference abundance and has better unmixing effect.

Data2 dataset is also composed of three datasets with 4, 5, and 6 endmembers, respectively. In Table V, we show the average eSAD and MSE of the endmembers and abundance map estimated using ten methods for each number of endmembers. Table V shows that in the unmixing results of three datasets with different endmembers number. The eSAD and MSE of the endmembers estimated by Rev-Net reach the minimum.

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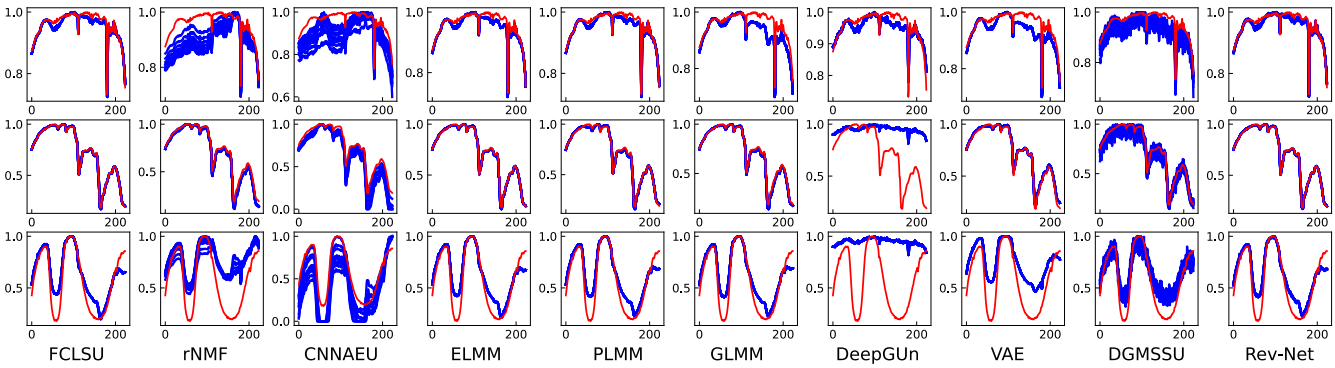


Fig. 9. Endmembers extracted from the synthetic DATA1-SNR = 30 db dataset using ten methods, each run ten times. The red curve in the figure represents three true endmembers selected from USGS spectral library, and the blue curve represents the endmembers estimated by ten algorithms at each time.

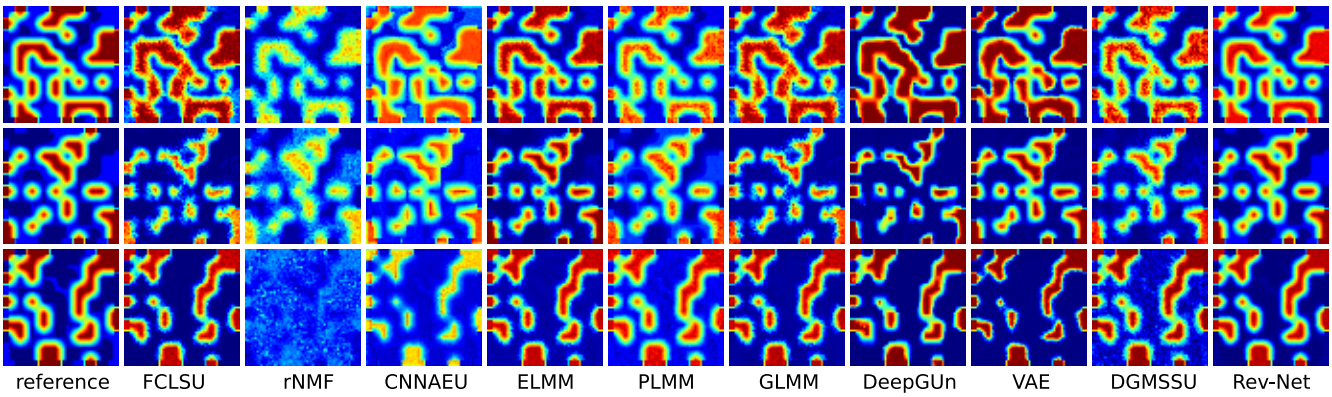


Fig. 10. Estimated abundance maps for the synthetic DATA1-SNR = 30 db. Abundance values represented by colors range from blue (0) to red (1).

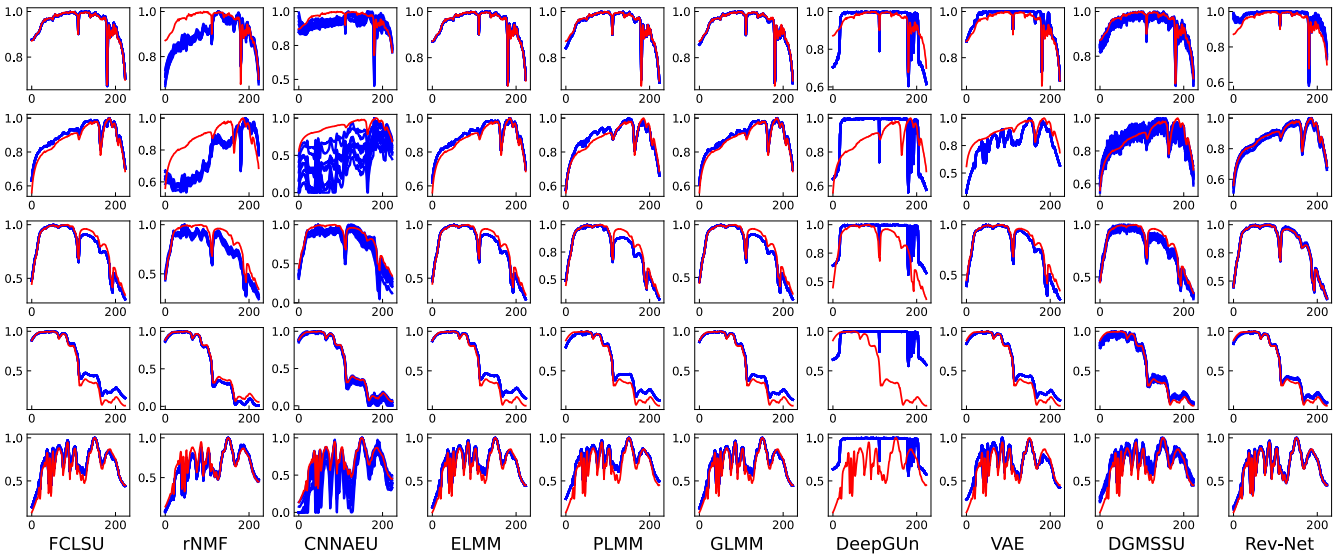


Fig. 11. Endmembers extracted from the synthetic DATA2-edm5 dataset using ten methods, each run ten times. The red curve in the figure represents five true endmembers selected from USGS spectral library, and the blue curve represents the endmembers estimated by ten algorithms at each time.

Rev-Net estimates the abundance of eSAD on three datasets. As shown in the table, when the number of endmembers increases, the estimation accuracy of various methods for endmembers and corresponding abundance decreases relatively. However, for endmembers and abundance map, the difference between estimation of Rev-Net and reference decreases relatively slowly, which also demonstrates the competitiveness of our method.

We provide the average eSAD and the average MSE of corresponding abundance for each endmember of the three datasets, respectively. Table VI shows the experimental results of four endmembers, showing that the MSE of the abundance

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TABLE V

eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SYNTHETIC DATA2 (10^{-2})

data	Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
edm4	eSAD	5.46	5.81±0.24	18.42±8.46	5.37	5.27	5.00	29.84	6.11	6.65±0.31	**3.70**±0.35
	MSE	5.31	22.14±0.26	12.46±4.20	3.18	3.65	5.10	9.95	4.63	5.44±0.53	**1.10**±0.30
edm5	eSAD	4.49	8.27±0.31	13.64±5.28	4.37	5.23	4.02	22.88	6.71	5.17±0.24	**2.32**±0.17
	MSE	8.01	27.53±0.21	20.35±7.08	5.32	9.69	7.49	56.76	20.32	14.16±3.05	**2.75**±0.24
edm6	eSAD	6.82	14.10±0.12	21.39±11.45	6.84	10.30	6.81	29.04	8.52	7.56±0.24	**4.73**±0.13
	MSE	24.50	33.78±0.36	27.02±12.81	22.90	18.81	25.34	70.02	19.31	9.50±0.98	**3.43**±0.21

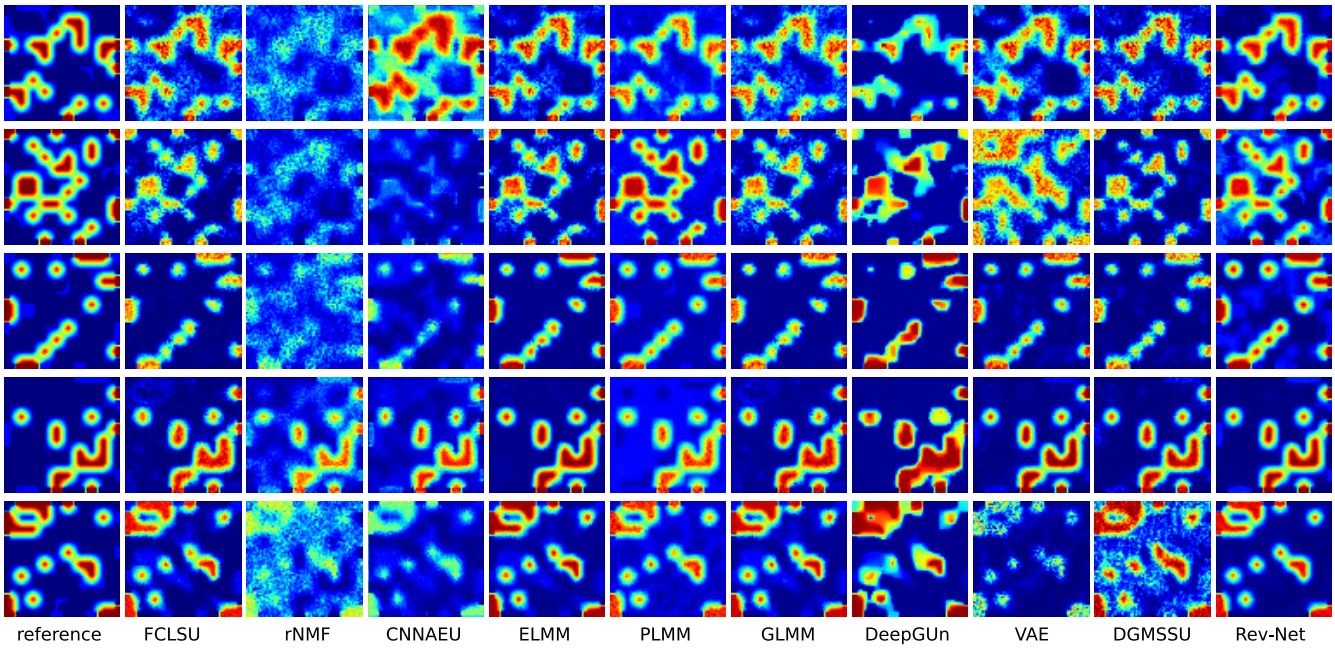


Fig. 12. Estimated abundance maps for the synthetic DATA2-edm5. Abundance values represented by colors range from blue (0) to red (1).

TABLE VI

eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SYNTHETIC DATA2-EDM4 (10^{-2})

endmember	Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
1st	eSAD	6.07	6.38±0.20	27.77±21.75	5.96	5.86	5.74	31.34	**3.76**	6.74±0.33	4.19±0.21
	MSE	1.36	5.74±0.11	2.96±1.06	0.71	0.62	1.27	2.02	1.00	1.25±0.19	**0.21**±0.06
2nd	eSAD	9.16	**3.48**±0.06	12.41±8.49	9.06	9.04	7.74	48.31	12.62	10.91±0.47	8.57±1.48
	MSE	0.53	3.53±0.02	1.91±1.05	0.44	1.26	0.51	3.97	0.53	0.60±0.07	**0.30**±0.06
3rd	eSAD	1.56	2.61±0.56	6.87±1.61	1.49	1.50	1.44	7.45	1.12	2.39±0.21	**0.93**±0.15
	MSE	2.03	9.59±0.06	4.61±1.34	1.27	1.01	1.97	1.97	2.39	2.52±0.34	**0.52**±0.18
4th	eSAD	5.06	10.78±0.93	26.62±6.99	4.95	4.69	5.06	32.24	6.95	6.58±0.45	**1.12**±0.29
	MSE	1.40	3.28±0.15	2.99±1.60	0.76	0.76	1.36	1.98	0.71	1.06±0.08	**0.07**±0.01

TABLE VII

eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SYNTHETIC DATA2-EDM5 (10^{-2})

endmember	Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
1st	eSAD	0.41	5.90±0.61	4.05±0.76	**0.38**	0.72	0.49	6.98	1.22	1.35±0.11	2.44±0.34
	MSE	3.10	6.22±0.11	4.59±2.65	1.72	0.98	3.02	1.48	2.43	2.87±0.41	**0.98**±0.13
2nd	eSAD	1.84	15.10±0.70	35.30±18.75	1.66	2.83	1.45	13.01	9.17	5.02±0.48	**0.69**±0.16
	MSE	2.86	9.23±0.07	7.25±2.34	2.17	2.74	2.59	3.19	9.92	4.78±1.56	**1.04**±0.12
3rd	eSAD	3.75	6.66±0.82	7.21±2.64	3.75	4.13	4.06	14.76	5.20	4.08±0.23	**0.62**±0.16
	MSE	0.85	4.89±0.13	3.60±1.67	0.34	1.78	0.78	25.01	1.09	1.34±0.18	**0.17**±0.06
4th	eSAD	10.21	5.87±0.07	**4.27**±2.36	10.02	10.35	8.51	55.41	8.66	5.70±0.31	5.27±0.43
	MSE	0.54	1.44±0.01	0.94±0.46	0.57	2.94	0.51	2.66	0.21	0.21±0.02	**0.18**±0.01
5th	eSAD	6.24	7.79±0.37	17.35±8.50	6.03	8.13	5.58	24.26	9.27	9.68±0.52	**2.61**±0.37
	MSE	0.65	5.75±0.11	3.97±1.66	0.52	1.24	0.60	24.42	6.67	4.96±1.41	**0.37**±0.04

matrix corresponding to all endmembers reaches the minimum, the eSAD of the first endmember estimated by VAE is the minimum. The eSAD of the second endmember estimated by rNMF is the minimum, and the eSAD of the last two

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TABLE VIII

eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SYNTHETIC DATA2-EDM6 (10^{-2})

endmember	Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
1st	eSAD	5.59	**3.53**±0.26	4.27±0.83	5.57	6.34	5.43	36.30	5.28	7.70±0.69	4.70±0.24
	MSE	0.57	5.26±0.10	3.88±1.61	0.48	1.27	0.53	16.59	0.52	0.90±0.33	**0.22**±0.02
2nd	eSAD	17.30	5.49±0.45	6.36±4.00	17.27	24.27	17.24	21.08	3.02	**1.78**±0.18	9.23±0.60
	MSE	9.63	5.51±0.06	7.32±8.62	9.76	4.39	9.86	8.57	7.01	2.78±0.35	**0.97**±0.08
3rd	eSAD	**0.93**	26.95±0.58	18.30±36.02	1.38	1.80	1.80	19.14	5.92	3.36±0.32	1.61±0.27
	MSE	11.00	5.04±0.09	4.41±1.19	9.85	9.92	11.76	17.18	7.94	2.77±0.31	**1.07**±0.08
4th	eSAD	1.90	9.26±0.73	20.36±7.38	1.81	3.23	1.69	1.45	**1.31**	5.19±0.56	2.30±0.32
	MSE	0.58	4.98±0.10	3.86±2.23	0.37	1.00	0.53	10.68	1.18	0.89±0.10	**0.29**±0.02
5th	eSAD	11.08	25.42±0.45	65.26±37.72	11.04	19.49	11.01	47.81	27.67	15.42±0.83	**7.56**±0.49
	MSE	2.39	8.71±0.18	5.07±3.55	2.27	1.18	2.38	3.92	2.10	1.02±0.11	**0.75**±0.07
6th	eSAD	4.10	13.97±0.39	13.80±7.77	3.97	6.66	3.68	48.45	7.95	11.92±0.67	**2.96**±0.37
	MSE	0.33	4.28±0.13	2.48±0.66	0.17	1.06	0.29	13.07	0.56	1.15±0.60	**0.14**±0.01

TABLE IX

TIME COST OF REV-NET ON REAL AND SYNTHETIC DATASET (10^{-2})

DATA	DATA1-db20	DATA1-db30	DATA1-db40	DATA2-edm4	DATA2-edm5	DATA2-edm6
time	41'4"	40'8"	40'57"	90'33"	114'2"	136'41"

TABLE X

ABLATION ANALYSIS ON THE SYNTHETIC DATA2 WITH eSAD AND MSE(10^{-2})

Method	edm4		edm5		edm6	
	eSAD	MSE	eSAD	MSE	eSAD	MSE
Rev-Net	**3.70**±0.35	**1.10**±0.30	**2.32**±0.17	2.75±0.24	**4.73**±0.13	**3.43**±0.21
without EM learning module	32.95±0	115.67±20.11	24.84±0	109.31±24.40	26.92±0	122.25±17.06
without 1st stage of L_{edm}	32.92±0.09	98.13±24.82	7.57±5.05	15.48±22.89	26.89±0.04	107.40±16.68
without 2nd stage of L_{edm}	5.47±0.03	1.69±0.07	4.50±0.01	**2.44**±0.09	6.81±0.02	13.44±0.22

TABLE XI

eSAD AND MSE COMPARISON OF UNMIXING RESULTS ON SYNTHETIC DATA3 (10^{-2})

data	Index	FCLSU	rNMF	CNNAEU	ELMM	PLMM	GLMM	DeepGUn	VAE	DGMSSU	Rev-Net
edm4	eSAD	3.86	13.23±0.42	25.90±5.52	4.25	15.81	5.88	15.77	4.77	4.10±0.51	**3.32**±0.09
	MSE	14.04	38.04±0.47	24.96±6.23	11.95	15.93	13.71	82.00	11.48	5.92±0.73	**3.86**±1.06
edm5	eSAD	4.22	16.41±0.33	22.18±7.64	4.12	4.45	4.16	29.16	6.11	5.48±0.15	**2.80**±0.07
	MSE	2.44	29.02±1.52	11.32±2.42	**1.06**	3.87	2.15	64.19	2.32	1.57±0.42	1.86±0.23
edm6	eSAD	2.56	11.61±0.43	21.32±4.49	2.52	3.66	2.42	15.13	3.47	5.64±0.90	**2.24**±0.10
	MSE	8.44	29.82±0.17	24.79±4.89	6.57	9.86	8.60	94.62	14.87	8.94±5.05	**2.14**±0.21

endmembers estimated by Rev-Net is the minimum. Table VII shows the experimental results of five endmembers, in which the MSE values of all endmembers estimated by Rev-Net reach the minimum compared with other methods. The first endmember obtained by ELMM has the smallest eSAD, the fourth endmember obtained by CNNAUE has the smallest eSAD, and the eSAD of the other endmembers estimated by the Rev-Net reaches the smallest. Table VIII shows the experimental results of six endmembers, showing that the MSE of the abundance matrix corresponding to all endmembers reaches the minimum. The methods for the first three endmembers to reach the minimum value of eSAD are rNMF, VAE, and VCA, respectively, while the method for the last three endmembers to reach the minimum value of eSAD is Rev-Net. In general, our method still obtains a good unmixing effect.

At the same time, we also provide the visualization results of the endmember and abundance maps obtained by various methods after unmixing datasets with different endmembers. Fig. 11 shows the ten runs results of the various methods on five endmembers, respectively, where Rev-Net all show relatively good experimental results. For example, the third and fourth endmembers in Fig. 11 intuitively demonstrate our relatively better experimental results. Fig. 12 shows the abundance map estimation of five endmembers by various methods, respectively. Rev-Net can restore the shape of the abundance map to a greater extent and make the overall color is purer, as well as the abundance map results are closer to the reference abundance and more reliable. What is more, we put the time of the experiment run in the Table IX, which was run on NVIDIA GeForce RTX 3090.

D. Ablation Study

In order to verify the effectiveness of each part of the proposed algorithm, we conducted ablation experiment. First, the main module proposed in this article is the endmember learning module. In order to verify the effectiveness of this module, we show that the endmember learning module is replaced by an endmember matrix, and the loss function is

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set as L_{rec}, L_{sparse} and L_{edm}, where L_{edm} only includes the first stage of L_{edm} proposed in Section II-F. Experimental results are shown in Table X. Second, in order to prove the necessity of dividing L_{edm} into two stages, we show the experimental results of removing the first and second stages of L_{edm}, respectively, which are also shown in Table X. From the table, we can find that the various components are necessary.

In addition, our algorithm can also work on datasets without spectral variability. To prove this, we set the dataset DATA3 with no spectral variability. Through the experimental results shown in Table XI, we can find that our method still shows good results on the synthetic dataset without spectral variability.

IV. Conclusion

In order to address the problem of spectral variability in hyperspectral unmixing, we propose a model based on reversible generative modules in this article. The model involves the endmember learning module and the abundance guidance module. In the endmember learning module, we regard the endmember as a random variable subject to unknown distribution and utilize the reversible module to get the explicit expression of the distribution, so as to quantitatively describe the spectral variability of hyperspectral images, and finally achieve the purpose of improving the unmixing effect. In the part of experimental, we carry out experiments on hyperspectral real dataset and synthetic datasets, respectively, and compare Rev-Net with nine other methods. The experimental results show that our method preserves competitive unmixing performance.

In the future, we will attempt to achieve a good unmixing effect without using the extracted endmembers as a priori, so as to avoid the impact of initialization.

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